

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1205

John Day 2 138 kV
September 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1205. The Interconnection Customer proposes a 79.2 MW battery energy storage facility interconnecting at the John Day 2 138 kV substation in the Pennsylvania area. The request was received on May 15, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 405 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$136,434,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sulphur Springs – Hilltop 138kV
Contingency: Strawberry Belt – Hilltop 345kV L
Loading: 198.1% (Rating: 302 MVA)
DFAX: 8.4%
Pre-project loading: 195.9%
2. Facility: Hilltop 345-138 kV XFMR
Contingency: Hilltop – Strawberry Belt 345kV
Loading: 191.8% (Rating: 389 MVA)
DFAX: 9.5%
Pre-project loading: 189.9%
3. Facility: Panda Patriot Generation Plant – Finley Butte 138kV
Contingency: Talenenergy Martins Creek Llc Williamsport – Montour Sta 345kV Line
Loading: 152.0% (Rating: 185 MVA)
DFAX: 26.7%
Pre-project loading: 140.5%
4. Facility: John Day 2 Substation – Talenenergy Martins Crk Llc Williamsport 138kV
Contingency: Montour Sta – Talenenergy Martins Crk Llc Williamsport 345kV
Loading: 140.3% (Rating: 297 MVA)
DFAX: 60.2%
Pre-project loading: 124.3%
5. Facility: Talenenergy Martins Crk Llc Williamsport 345-138 kV Transformer Bank
Contingency: Montour Station – Talenenergy Martins Creek Llc Williamsport 345kV line
Loading: 135.5% (Rating: 422 MVA)
DFAX: 35.0%
Pre-project loading: 129.0%

6. Facility: Rock Spring – Finley Butte 138kV
Contingency: Montour Station – Talenenergy Martins Crk Llc Williamsport 345kV
Loading: 127.4% (Rating: 206 MVA)
DFAX: 26.7%
Pre-project loading: 117.1%
7. Facility: Talenenergy Martins Crk Llc Lock Haven Substation – El Cerrito Sta G 138kV Line
Contingency: American Crystal Sugar Drayton – Scott 345kV
Loading: 126.4% (Rating: 235 MVA)
DFAX: 7.3%
Pre-project loading: 123.9%
8. Facility: John Day 2 – Panda Patriot Generation Plant 138kV
Contingency: Montour Station – Talenenergy Martins Crk Llc Williamsport 345kV
Loading: 124.6% (Rating: 225 MVA)
DFAX: 27.5%
Pre-project loading: 114.9%
9. Facility: Pastillo 5801 – Taylor Flts 138kV L
Contingency: Montour Sta – Talenenergy Martins Creek Llc Williamsport 345kV
Loading: 121.4% (Rating: 202 MVA)
DFAX: 5.7%
Pre-project loading: 119.2%
10. Facility: Dairyville – Luverne Wind Farm 138kV
Contingency: Strawberry Belt – Jh Warden 345kV
Loading: 120.4% (Rating: 216 MVA)
DFAX: 5.3%
Pre-project loading: 118.5%
11. Facility: Coopervision – Power Inn 138kV
Contingency: Montour Station – Talenenergy Martins Creek Llc Williamsport 345kV
Loading: 119.5% (Rating: 225 MVA)
DFAX: 5.2%
Pre-project loading: 117.7%
12. Facility: Lawnwood – Sulphur Springs 138kV
Contingency: Strawberry Belt – Hilltop 345kV
Loading: 112.3% (Rating: 297 MVA)
DFAX: 6.5%
Pre-project loading: 110.6%
13. Facility: Talenenergy Martins Creek Llc Lock Haven – Taylor Flats 138kV
Contingency: American Crystal Sugar Drayton – Scott 345kV
Loading: 112.0% (Rating: 265 MVA)
DFAX: 7.3%
Pre-project loading: 109.8%
14. Facility: El Cerrito Sta G Substation 138 kV/1 kV Transformer
Contingency: Shawville Sta – Scott 345kV Line
Loading: 111.7% (Rating: 325 MVA)
DFAX: 6.2%
Pre-project loading: 110.2%
15. Facility: Blackrock – Ekluna 345kV
Contingency: Caldwell – Piney Crk Project 345kV
Loading: 103.2% (Rating: 1,380 MVA)
DFAX: 6.4%
Pre-project loading: 102.8%
16. Facility: E Fayetteville – El Dorado 138kV
Contingency: John Day 2 – Talenenergy Martins Crk Llc Williamsport 138kV

Loading: 101.5% (Rating: 192 MVA)
 DFAX: 5.6%
 Pre-project loading: 99.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hilltop – Sulphur Springs 138kV	Rebuild line with higher-capacity conductor	\$4,159,000	100% (\$4,159,000)
Hilltop 345/138kV Auto	Replace transformer with higher-rated unit	\$12,686,000	100% (\$12,686,000)
Panda Patriot Generation Plant – Finley Butte 138kV	Rebuild line with higher-capacity conductor	\$15,141,000	100% (\$15,141,000)
John Day 2 – Talenenergy Martins Crk Llc Williamsport 138kV	Rebuild line with higher-capacity conductor	\$9,379,000	100% (\$9,379,000)
Talenenergy Martins Crk Llc Williamsport 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$10,979,000	100% (\$10,979,000)
Rock Spring – Finley Butte 138kV	Rebuild line with higher-capacity conductor	\$2,876,000	100% (\$2,876,000)
Talenenergy Martins Crk Llc Lock Haven Substation – El Cerrito Sta G 138kV Line	Rebuild line with higher-capacity conductor	\$13,336,000	100% (\$13,336,000)
John Day 2 – Panda Patriot Generation Plant 138kV	Rebuild line with higher-capacity conductor	\$5,356,000	100% (\$5,356,000)
Taylor Flats – Pastillo 5801 138kV	Rebuild line with higher-capacity conductor	\$27,464,000	100% (\$27,464,000)
Dairyville – Luverne Wind Farm 138kV	Rebuild line with higher-capacity conductor	\$4,922,000	100% (\$4,922,000)
Coopervision – Power Inn 138kV Line	Rebuild line with higher-capacity conductor	\$3,534,000	100% (\$3,534,000)
Lawnwood – Sulphur Springs 138kV	Rebuild line with higher-capacity conductor	\$14,039,000	100% (\$14,039,000)
Talenenergy Martins Creek Llc Lock Haven – Taylor Flats 138kV	Rebuild line with higher-capacity conductor	\$5,592,000	100% (\$5,592,000)
El Cerrito Sta G Substation 138 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$4,976,000	100% (\$4,976,000)
Eklutna – Blackrock 345kV	Reconductor line	\$27,037,000	2% (\$657,000)
E Fayetteville – El Dorado 138kV	Reconductor line	\$1,338,000	100% (\$1,338,000)
	Total allocated to project		\$136,434,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.